

Causal Survival Analysis: Practical Recommendations

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Causal Survival analysis: example of questions

Survival analysis



Causal inference

⇒ **Effect of a policy/intervention/treatment A on an time to event outcome T**



What is the impact of an oncology medicine on long term mortality ?



What is the impact of a medicine all along the time (it can be beneficial in the short term but harmful in the long term) ?



Treatment : A

Treatment effect



Time to event : T

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Survival analysis



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Treatment : A

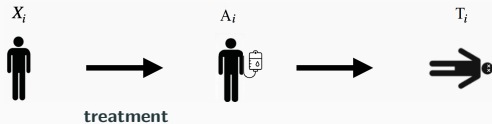
Treatment effect



Time to event : T



Potential outcome framework (D. B. Rubin 1974)



$$\Rightarrow n \text{ i.i.d. } (\underbrace{X_i}_{\text{covariates}}, \underbrace{A_i}_{\text{treatment}}, \underbrace{T_i}_{\text{outcome}}) \in \mathbb{R}^d \times \{0, 1\} \times \mathbb{R} \times \mathbb{R}$$

Let's say that in our example $X_1 = \text{sex}$ and $X_2 = \text{age}$.

Covariates		Treatment	Censoring	Status	Outcomes		
X_1	X_2	A	C	Δ	$T(0)$	$T(1)$	$\tilde{T}$
1	24	1	?	1	?	200	200
2	52	0	?	1	100	?	100
1	33	1	200	0	?	?	200

In grey, the observed data: $(X_i, A_i, \Delta_i, \tilde{T}_i)$ with $\tilde{T}_i = \min(T_i, C_i)$

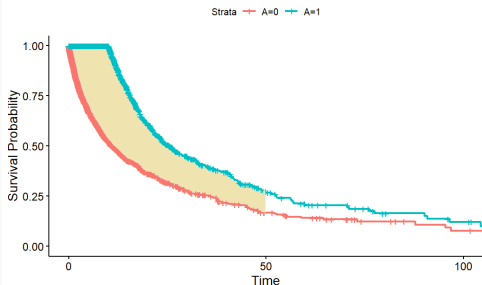
S.T.U.V.A. $T = AT(1) + (1 - A)T(0) \Rightarrow T$ is not directly observed

Causal effect in survival analysis

Difference in Restricted Mean Survival Time (RMST): Average treatment effect in survival analysis

$$\begin{aligned}\hat{\theta}_{RMST}(\tau) &= E[\min(T(1), \tau) - \min(T(0), \tau)] \\ &= \int_0^{\tau} (\hat{S}_1(t) - \hat{S}_0(t)) dt\end{aligned}$$

RMST can be defined as a measure of average survival from time 0 to time τ a **fixed time horizon**



$\hat{\theta}_{RMST}(\tau = 50) = 10$ means that on average the treatment increases the survival time by 10 days at 50 days.

Figure 1: Plot of stratified kaplan meier survival function and the representation of $\theta_{RMST}(\tau = 50)$ (in yellow)

Identifiability assumptions

$$\text{S.T.U.V.A. } T = AT(1) + (1 - A)T(0)$$

RCT & Independent censoring

- **Random treatment assignment**
 $A \perp\!\!\!\perp (T(0), T(1), X)$
- **Trial positivity** $0 < \mathbb{P}(A = 1) < 1$
- **Independent censoring**
 $C \perp\!\!\!\perp (T(0), T(1), X, A)$

RCT & Dependent censoring

- **Random treatment assignment**
 $A \perp\!\!\!\perp (T(0), T(1), X)$
- **Conditionally independent censoring**
 $C \perp\!\!\!\perp (T(0), T(1)) | X, A$
- **Positivity for censoring**
 $\mathbb{P}(C \geq t | X = x, A = a) \geq \varepsilon$

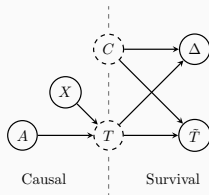
Obs & Independent censoring

- **Unconfoundedness**
 $A \perp\!\!\!\perp (T(0), T(1)) | X$
- **Positivity for treatment**
 $0 < \mathbb{P}(A = 1 | X) < 1$
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RCT & Independent censoring



Non-adjusted Kaplan-Meier estimator [Kaplan and Meier 1958]

It corresponds to a simple Kaplan meier estimator:

$$\hat{S}_{KM}(t|A=a) := \prod_{t_k \leq t} \left(1 - \frac{D_k(a)}{N_k(a)}\right).$$

with (D_k) the number of deaths and (N_k) the individuals at risk at time t_k in the treated group ($a=1$) or in the control group ($a=0$)

Asymptotic property: $\sqrt{n} \left(\hat{S}_{KM}(t|A=a) - S^{(a)}(t) \right) \xrightarrow{d} \mathcal{N}(0, V_{KM})$

$$\hat{\theta}_{RMST}(\tau) = \int_0^\tau \left(\hat{S}_1(t) - \hat{S}_0(t) \right) dt$$

Identifiability assumptions

$$\text{S.T.U.V.A. } T = AT(1) + (1 - A)T(0)$$

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- **Conditionally independent censoring**
 $C \perp\!\!\!\perp (T(0), T(1)) | X, A$
- **Positivity for censoring**
 $\mathbb{P}(C \geq t | X, A) = G(t | X, A) \geq \varepsilon$

Obs & Independent censoring

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- **Positivity for treatment**
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RCT & Dependent censoring

Censoring unbiased transformation T^* [Fan and Gijbels 1994]:

The basic requirement is that $E(T^*|X, A) = E(T \wedge \tau|X, A)$.

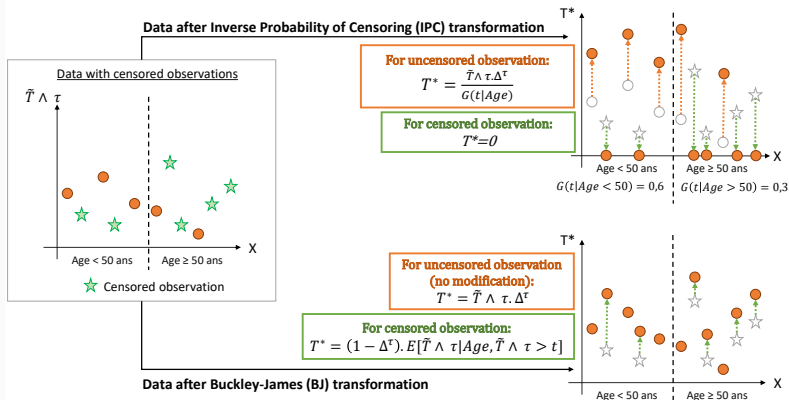


Figure 2: Illustration of Inverse-Probability-of-Censoring and Buckley-James transformations

RCT & Dependent censoring

Censoring unbiased transformation T^* [Fan and Gijbels 1994]:

The basic requirement is that $E(T^*|X, A) = E(T \wedge \tau|X, A)$.

- **IPC transformation** [Koul, Susarla, and Ryzin 1981]:

$$T_{\text{IPCW}}^* = \frac{\Delta^\tau}{G(\tilde{T} \wedge \tau|X, A)} \tilde{T} \wedge \tau$$

with $G(\tilde{T} \wedge \tau|X, A) := \mathbb{P}(C \geq t|X, A)$ is the left limit of the **conditional survival function of the censoring**. $\Delta^\tau := I\{T \wedge \tau \leq C\}$ is the censoring indicator of the restricted time (or restricted status).

- **Buckley-James transformation** [Buckley and James 1979]:

$$T_{\text{BJ}}^* = \Delta^\tau(\tilde{T} \wedge \tau) + (1 - \Delta^\tau)Q_S(\tilde{T} \wedge \tau|X, A)$$

where, for $t \leq \tau$, $Q_S(t|X, A) := \mathbb{E}[T \wedge \tau|X, A, T \wedge \tau > t]$ estimated with the **conditional survival function** $S(t|X, A = a)$.

RCT & Dependent censoring

IPCW adjusted Kaplan-Meier estimator [Robins and Finkelstein 2000]

$$\hat{S}_{\text{IPCW}}(t|A=a) = \prod_{t_k \leq t} \left(1 - \frac{D_k^{\text{IPCW}}(a)}{N_k^{\text{IPCW}}(a)} \right).$$

with $D_k^{\text{IPCW}}(a) := \sum_{i=1}^n \frac{\Delta_i^\tau}{\hat{G}(\tilde{T}_i \wedge \tau | X_i, A=a)} I(\tilde{T}_i = t_k, A_i = a)$ and

$$N_k^{\text{IPCW}}(a) := \sum_{i=1}^n \frac{\Delta_i^\tau}{\hat{G}(\tilde{T}_i \wedge \tau | X_i, A=a)} I(\tilde{T}_i \geq t_k, A_i = a)$$

Every uncensored observation is weighted by the inverse of the probability of remain uncensored given the covariates.

Asymptotic property: If $\hat{G}$ is uniformly weakly (resp. strongly) consistent then $\hat{S}_{\text{IPCW}}(t|A=a)$ is uniformly weakly (resp. strongly) consistent and so is $\hat{\theta}_{\text{IPCW-KM}}$.

$$\hat{\theta}_{\text{IPCW-KM}} = \int_0^\tau \hat{S}_{\text{IPCW}}(t|A=1) - \hat{S}_{\text{IPCW}}(t|A=0) dt.$$

RCT & Dependent censoring

Buckley-James based estimator

$$\hat{\theta}_{\text{BJ}} = \frac{1}{n_1} \sum_{A_i=1} T_{\text{BJ},i}^* - \frac{1}{n_0} \sum_{A_i=0} T_{\text{BJ},i}^*$$

- n_1 corresponds to the number of observations in the treated group.
- n_2 corresponds to the number of observations in the control group.
- $T_{\text{BJ}}^* = \Delta^\tau(\tilde{T} \wedge \tau) + (1 - \Delta^\tau)Q_S(\tilde{T} \wedge \tau|X, A)$ the estimation function of the remaining survival function.

Asymptotic property: If $\hat{Q}_S$ is uniformly weakly (resp. strongly) consistent then so is $\hat{\theta}_{\text{BJ}}$.

Identifiability assumptions

$$\text{S.T.U.V.A. } T = AT(1) + (1 - A)T(0)$$

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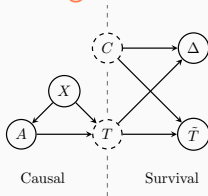
Obs & Independent censoring

- Unconfoundedness
 $A \perp\!\!\!\perp (T(0), T(1)) | X$
- Positivity for treatment
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Obs & Independent censoring



IPTW adjusted Kaplan-Meier estimator [Xie and Liu 2005]

$$\hat{S}_{\text{IPTW}}(t|A=a) = \prod_{t_k \leq t} \left(1 - \frac{D_k^{\text{IPTW}}(a)}{N_k^{\text{IPTW}}(a)} \right).$$

with $D_k^{\text{IPTW}}(a) := \sum_{i=1}^n \left(\frac{a}{\hat{e}(X_i)} + \frac{1-a}{1-\hat{e}(X_i)} \right) I(\tilde{T}_i = t_k, \Delta_i = 1, A_i = a)$ and $N_k^{\text{IPTW}}(a) := \sum_{i=1}^n \left(\frac{a}{\hat{e}(X_i)} + \frac{1-a}{1-\hat{e}(X_i)} \right) I(\tilde{T}_i \geq t_k, A_i = a)$ and $e(X_i) = P(A_i = 1|X_i)$, the propensity score.

Asymptotic property: If $\hat{e}(\cdot)$ satisfies $\|\hat{e} - e\|_\infty \rightarrow 0$ almost surely (resp. in probability), then $\hat{S}_{\text{IPTW}}(t|A=a)$ is strongly (resp. weakly) consistent, then so is $\hat{\theta}_{\text{RMST}}$.

$$\hat{\theta}_{\text{RMST}} = \int_0^\tau \hat{S}_{\text{IPTW}}(t, A=1) - \hat{S}_{\text{IPTW}}(t, A=0) dt$$

Identifiability assumptions

$$\text{S.T.U.V.A. } T = AT(1) + (1 - A)T(0)$$

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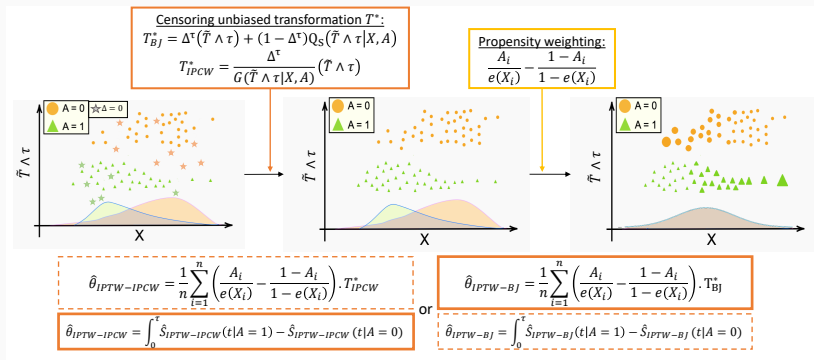
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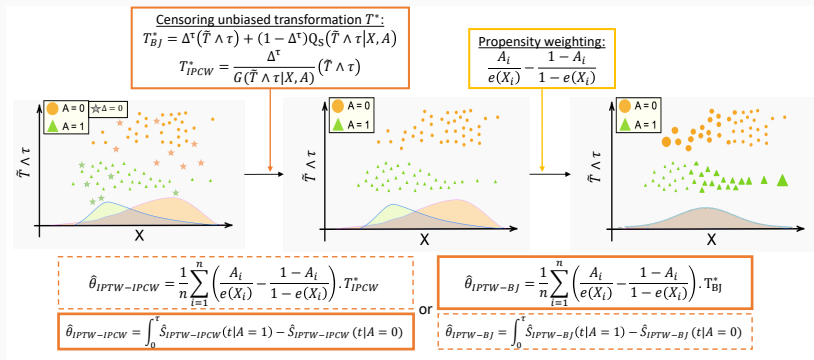
Obs & Dependent censoring



Asymptotic property:

- If $\hat{G}(\cdot|A, X)$ and $\hat{e}$ are weakly (resp. strongly) uniformly **consistent**, then $\hat{S}_{IPTW-IPCW}(t|A=a)$ is weakly (resp. strongly) consistent and **so is** $\hat{\theta}_{IPTW-IPCW}$.
- If $\hat{Q}_S(\cdot|X, A)$ and $\hat{e}(\cdot)$ are uniformly weakly (resp. strongly) **consistent** estimators, then $\hat{\theta}_{IPTW-BJ}$ is weakly (resp. strongly) consistent and **so is** $\hat{\theta}_{IPTW-BJ}$.

Obs & Dependent censoring



Asymptotic property:

- If $\hat{G}(\cdot|X)$ and $\hat{e}$ are weakly (resp. strongly) uniformly consistent, then $\hat{S}_{IPTW-IPCW}(t|A=a)$ is weakly (resp. strongly) consistent and so is $\hat{\theta}_{IPTW-IPCW}$.
- If $\hat{Q}_S(\cdot|X, A)$ and $\hat{e}(\cdot)$ are uniformly weakly (resp. strongly) consistent estimators, then $\hat{\theta}_{IPTW-BJ}$ is weakly (resp. strongly) consistent and so is $\hat{\theta}_{IPTW-BJ}$.

Obs & Dependent censoring

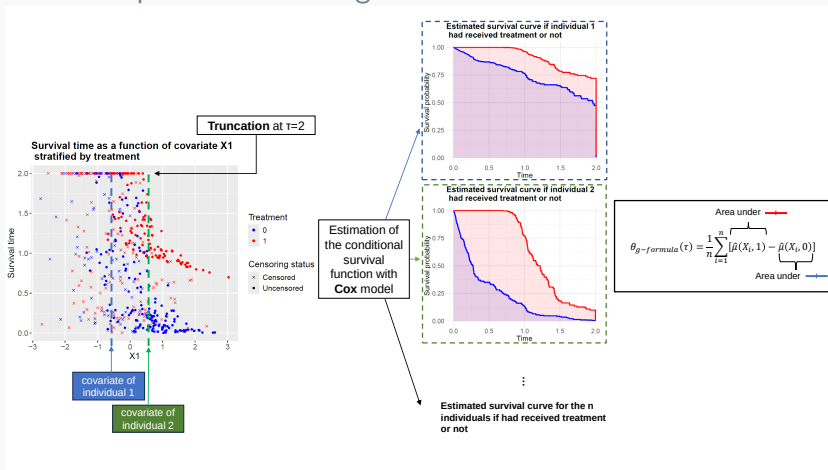
G-formula estimator

$$\hat{\theta}_{\text{G-formula}} = \frac{1}{n} \sum_{i=1}^n \hat{\mu}(X_i, 1) - \hat{\mu}(X_i, 0).$$

with $\hat{\mu}_a \triangleq \mathbb{E}[T \wedge \tau \mid X = x, A = a] = \int_0^\tau S(t \mid X = x, A = a) dt$ the integral of the **conditional survival function** truncated at τ .

- Predict the conditional survival for each individual based on their covariates and treatment → can be estimated by Cox model or survival forests
- $\theta_{\text{G-formula}}$ is estimated by averaging conditional survival estimates across all individuals and computing the difference between treated and control groups

Obs & Dependent censoring



Two possibilities for estimating $\hat{S}(t|x, a)$:

- S-learner: **Fit one model** on all data with adjustment on X and A.
- T-learner: **Fit two models** (stratified analysis) on data with A=1 and with A=0 and adjustment on X.

Obs & Dependent censoring

Another censoring unbiased transformation:

- **Doubly robust transformation for censoring** [D. Rubin and Laan 2007]:

$$T_{\text{DR}}^* = \frac{\Delta^\tau \tilde{T} \wedge \tau}{G(\tilde{T} \wedge \tau | X, A)} + \frac{(1 - \Delta^\tau) Q_S(\tilde{T} \wedge \tau | X, A)}{G(\tilde{T} \wedge \tau | X, A)} - \int_0^{\tilde{T} \wedge \tau} \frac{Q_S(t | X, A)}{G(t | X, A)^2} d\mathbb{P}_C(t | X, A),$$

- First term: IPC transformation
- Second term: BJ transformation with IPC Weighting
- Third term: Augmentation term

Obs & Dependent censoring

Augmented estimator: AIPTW-AIPCW [Ozenne et al. 2020]

$$\Delta_{\text{QR}}^*(\textcolor{brown}{G}, \textcolor{blue}{S}, \textcolor{blue}{\mu}, \textcolor{red}{e}) := \left(\frac{A}{\textcolor{red}{e}(X)} - \frac{1-A}{1-\textcolor{red}{e}(X)} \right) (T_{\text{DR}}^*(\textcolor{brown}{G}, \textcolor{blue}{S}) - \mu(X, A)) + \mu(X, 1) - \mu(X, 0)$$

$$\hat{\theta}_{\text{AIPTW-AIPCW}} := \frac{1}{n} \sum_{i=1}^n \Delta_{\text{QR}}^*(\hat{\textcolor{brown}{G}}, \hat{\textcolor{blue}{S}}, \hat{\textcolor{blue}{\mu}}, \hat{\textcolor{red}{e}}).$$

⇒ 3 nuisance parameters to compute :

- Censoring model : $C \sim A + X$
- Propensity score model : $A \sim X$
- Conditional survival : $T \sim A + X$

Asymptotic properties: Let F, R be two conditional survival function, p be a propensity score, and ν be a conditional response.

$$\mathbb{E}[\Delta_{\text{QR}}^*|X] = \mathbb{E}[T(1) \wedge \tau - T(0) \wedge \tau|X] \quad \text{if} \quad \begin{cases} F = G & \text{or} & R = S & \text{and} \\ p = e & \text{or} & \nu = \mu. \end{cases}$$

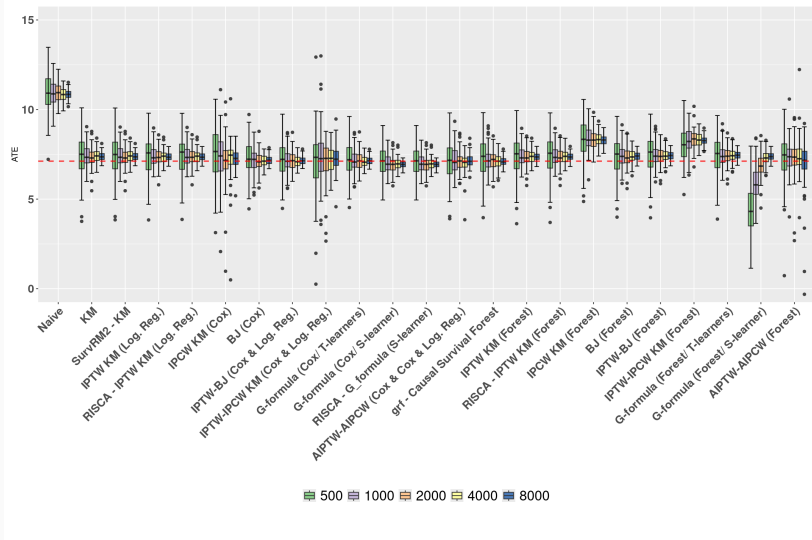
Obs & Dependent censoring

Estimator	Context of application	Outcome model	Censoring model	Treatment model	Robustness
<i>Unadjusted KM</i>	RCT & Indep. cens.				
IPCW-KM	RCT & Dep. cens.		(G)		No
BJ		(Q_S)			No
<i>IPTW-KM</i>	Obs. & Indep. cens.			(e)	No
IPCW-IPTW-KM	Obs & Dep. cens.		(G)	(e)	No
<i>G-formula</i>		(μ)			No
IPTW-BJ		(Q_S)		(e)	No
AIPTW-AIPCW		(Q_S, μ)	(G)	(e)	Yes

Table 1: Estimators of the difference in RMST and nuisance parameters needed to compute each estimator. Empty boxes indicate that the nuisance parameter is not needed in the estimator thus mis-specification has no impact. Estimators in italic are those that are already implemented in available packages.

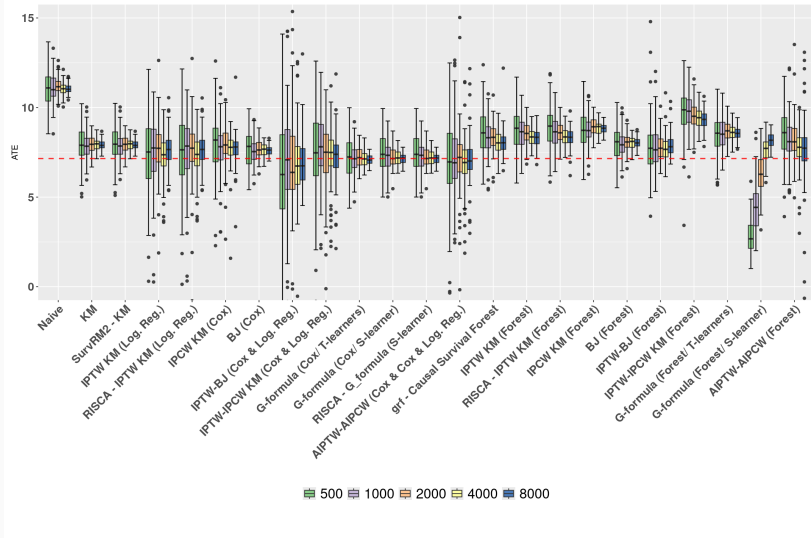
Consistency of the nuisance parameter estimators implies consistency of the RMST estimator except AIPTW-AIPCW estimator which requires either censoring and treatment or survival well specified

RCT & Dependent censoring ((semi)-parametric DGP)



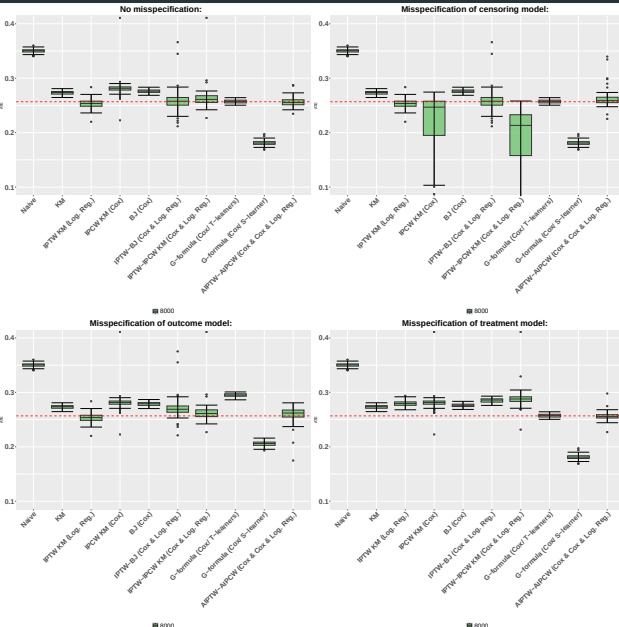
- Every estimators without correction for dependent censoring are biased.
- Best estimators: G-formula (Cox/ T-learners) and BJ (Cox).
- All forest estimators struggle to converge except AIPTW-AIPCW and Causal Survival Forest.

Obs & Dependent censoring ((semi)-parametric DGP)



- Every estimators without correction for dependent censoring and confounding bias are biased.
- Best estimators: G-formula (Cox/ T-learners).
- All forest estimators struggle to converge.

Misspecification: Obs & Dependent censoring



When there is no misspecification:

- IPTW-BJ (Cox Log.Reg), G-formula (Cox/T-learners) and AIPW-AIPCW (Cox Cox Reg.Log) are unbiased.

When there is misspecification of one nuisance parameter:

- AIPW-AIPCW (Cox Cox Reg.Log) is convergent when there is one nuisance misspecification.
- In contrary, the other estimators are biased when one of its nuisance parameter is misspecified.

Conclusions

- Difference in RMST is one interesting quantity to estimate average treatment effect in survival analysis.
- These estimators provide a valuable alternative to the Hazard Ratio.

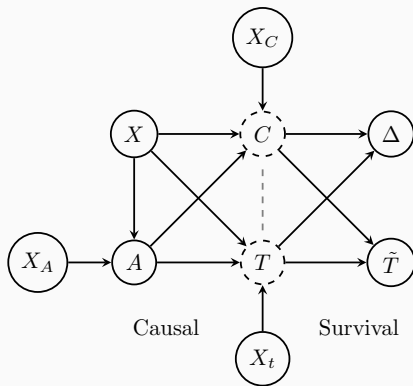
RECOMMENDATIONS

- IPCW very used in practice, instead we recommend implementing **BJ** which appears to have better statistical properties.
- **Causal Survival Forest, G-formula (T-learners), IPTW-BJ, and AIPTW-AIPCW** (the **doubly** robust estimator) in both their Cox, logistic regression and forest versions are very stable.

LIMITS

- Our simulations utilize large sample sizes with relatively simple relationships.

What's next ?



⇒ Focus on variable selection:

X is the minimum sufficient set but what happen if we include variable related only to the outcome or treatment or censoring ?

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Appendix

RCTs simulations

For the RCTs simulation, n samples $(X_i, A_i, C, T_i(0), T_i(1))$ are generated in the following way:

- $X \sim \mathcal{N}(\mu = [1, 1, -1, 1]^\top, \Sigma = I_4)$
- $e(X) = 0.5$ (constant) for the propensity score (A)
- $\lambda^{(0)}(t|X) = 0.01 \cdot \exp\{0.5X_1 + 0.5X_2 - 0.5X_3 + 0.5X_4\}$ hazard for the event time $T(0)$
- $T(1) = T(0) + 10$
- The hazard for the censoring time C :
 - Scenario 1: $\lambda_c = 0.03$.
 - Scenario 2:
 $\lambda_c(X) = 0.03 \cdot \exp\{0.7X_1 + 0.7X_2 - 0.25X_3 - 0.1X_4 - 0.2A\}$.
- The threshold time τ is set to 25.

Obs simulations

For the RCTs simulation, n samples $(X_i, A_i, C, T_i(0), T_i(1))$ are generated in the following way:

- $X \sim \mathcal{N}(\mu = [1, 1, -1, 1]^\top, \Sigma = I_4)$
- $\text{logit}(e(X)) = \beta_A^\top X$ where $\beta_A = (1, 1, 2.5, 1)$ ($\text{logit}(p) = \log(p/(1-p))$)
- $\lambda^{(0)}(t|X) = 0.01 \cdot \exp\{0.5X_1 + 0.5X_2 - 0.5X_3 + 0.5X_4\}$ hazard for the event time $T(0)$
- $T(1) = T(0) + 10$
- The hazard for the censoring time C :
 - Scenario 1: $\lambda_c = 0.03$.
 - Scenario 2:
 $\lambda_c(X) = 0.03 \cdot \exp\{0.7X_1 + 0.7X_2 - 0.25X_3 - 0.1X_4 - 0.2A\}$.
- The threshold time τ is set to 25.

Misspecified simulations

We generate n samples $(X_i, A_i, C, T_i(0), T_i(1))$ as follows:

$X \sim \mathcal{N}(\mu, \Sigma)$ and $\mu = (0.5, 0.5, 0.7, 0.5)$, $\Sigma = \text{Id}_4$

$\lambda^{(0)}(t|X) = \exp\{\beta_0^\top Y\}$ where $\beta_0 = (0.2, 0.3, 0.1, 0.1, 1, 0, 0, 0, 0, 1)$,
and $Y = (X_1^2, X_2^2, X_3^2, X_4^2, X_1X_2, X_1X_3, X_1X_4, X_2X_3, X_2X_4, X_3X_4)$.

$T(1) = T(0) + 1$.

$\lambda_C(t|X) = \exp\{\beta_C^\top Y\}$ where $\beta_C = (0.05, 0.05, -0.1, 0.1, 0, 1, 0, -1, 0, 0)$.

$\text{logit}(e(x)) = \beta_A^\top Y$ where $\beta_A = (0.05, -0.1, 0.5, -0.1, 1, 0, 1, 0, 0, 0)$.

When the model is well-specified, the full vector (X, Y) is given as an input of the nuisance parameter models. When it is not, only X and the first half of Y corresponding to $(X_1^2, X_2^2, X_3^2, X_4^2)$ is given as an input.